SOCIAL ENGINEERING ASSESSMENTS SHOULD BE MANDATORY DURING EVERY PENETRATION TEST

Team: A

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1. Introduction

Cybersecurity threats do not always target technology directly; they can also target people. Social engineering involves manipulating individuals into revealing confidential information or performing unsafe actions. A penetration test mainly identifies technical vulnerabilities, while a social engineering assessment checks how well employees can resist human-based attacks. Therefore, making social engineering assessments mandatory can provide a more complete picture of an organization’s security.

2. Definition

Social Engineering Assessment is a controlled security test that evaluates whether people in an organization can recognize and resist manipulation techniques such as phishing, impersonation, baiting, or pretexting.

Penetration Testing is an authorized security assessment in which ethical hackers simulate attacks against an organization's systems, networks, applications, or other assets to identify vulnerabilities.

3. Objectives

The main objectives of including social engineering assessments are:

- To identify weaknesses in employee security awareness.
- To test how employees respond to phishing and impersonation attempts.
- To identify human-related security vulnerabilities.
- To improve employee cybersecurity training.
- To reduce the possibility of successful real-world attacks.

4. Importance

Employees are often considered an important part of an organization's security system. Even if an organization has strong firewalls, antivirus

software, and access controls, an attacker may try to convince an employee to reveal credentials or open a malicious attachment. Social engineering assessments help organizations discover these weaknesses before real attackers exploit them.

5. Advantages

Making social engineering assessments part of every penetration test can provide several benefits:

- Early detection: Identifies human vulnerabilities before attackers exploit them.
- Better awareness: Helps employees understand real-world attack techniques.
- Improved security: Encourages safer handling of passwords, emails, and sensitive information.
- Realistic testing: Simulates attacks that employees may actually encounter.
- Reduced risk: Helps prevent data breaches caused by human mistakes.
- Complete assessment: Tests both technical and human aspects of security.

6. Example Situation

Suppose an ethical hacker sends an authorized simulated phishing email to employees pretending to be from the organization's IT department. The email asks employees to verify their account through a test website. If several employees enter their credentials, the assessment shows that the organization has a weakness in phishing awareness.

7. Arguments for Making It Mandatory

Social engineering assessments should be mandatory because technical security alone cannot protect an organization from attacks that exploit human behavior. Attackers frequently use methods such as phishing, impersonation, and manipulation to bypass security controls. Including these assessments in every penetration test ensures that organizations evaluate their people, processes, and technology, rather than focusing only on technical systems. However, the assessments should always be authorized, controlled, ethical, and designed to avoid unnecessary harm or embarrassment to employees.

8. Conclusion

In conclusion, social engineering assessments can reveal security weaknesses that traditional technical penetration testing may miss. Since employees can become targets of phishing, impersonation, and other manipulation techniques, testing human security is an important part of understanding an organization's overall risk. Therefore, social engineering assessments should be included as a mandatory component of penetration testing, with proper authorization, clear boundaries, and employee-focused security training.


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